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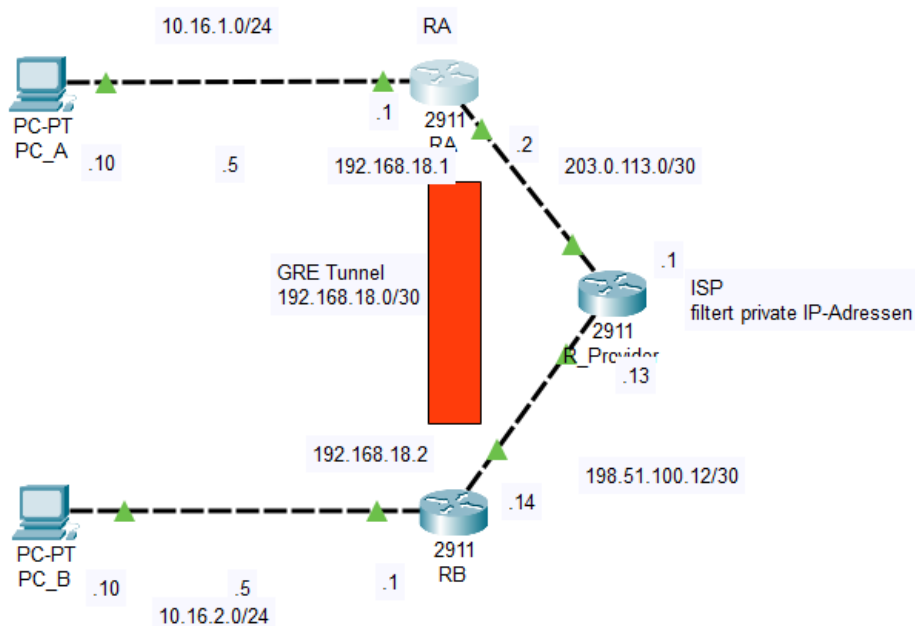
Thema der Laborübung:

Konfiguration eines Point-to-Point GRE-VPN-Tunnels mit statischen Routen

1. Ziel der Übung

Ziel ist es, die beiden internen Netzwerke 10.16.1.0/24 (Standort A) und 10.16.2.0/24 (Standort B) über das Internet zu verbinden. Da der Provider private IP-Adressen filtert und NAT nicht verwendet werden soll, wird ein GRE-Tunnel zwischen RA und RB aufgebaut. Das Routing erfolgt ausschließlich über statische Routen (kein OSPF).

Topologie: (hier KN=18)



Gerät	Interface	IP Adresse	Subnetz Maske	Default Gateway
RA	G0/0	10.16.1.1	255.255.255.0	-
	G0/1	203.0.113.2	255.255.255.252	-
	Tunnel0	192.168.18.1	255.255.255.252	-
ISP	G0/0	203.0.113.1	255.255.255.252	-
	G0/1	198.51.100.13	255.255.255.252	-
RB	G0/0	10.16.2.1	255.255.255.0	-
	G0/1	198.51.100.14	255.255.255.252	-
	Tunnel0	192.168.18.2	255.255.255.252	-
PC-A	FastEthernet0	10.16.1.10	255.255.255.0	10.16.1.1
PC-B	FastEthernet0	10.16.2.10	255.255.255.0	10.16.2.1

2. Konfiguration der Router RA und RB

Zuerst wurden beide Router (RA und RB) vollständig konfiguriert.

Dabei wurden die Standard-Geräteeinstellungen gemäß Angabe vorgenommen: DNS-Lookup wurde deaktiviert, die Hostnamen (RA und RB) gesetzt, die Passwortverschlüsselung aktiviert sowie die Passwörter festgelegt (Console/VTY: cisco, Enable Secret: class).

Anschließend wurden die IP-Adressen auf den Interfaces konfiguriert und aktiviert:

RA:

- G0/0 → 10.16.1.1/24
- G0/1 → 203.0.113.2/30

RB:

- G0/0 → 10.16.2.1/24
- G0/1 → 198.51.100.14/30

Danach wurden auf beiden Routern statische Routen gesetzt.

Jeder Router erhielt:

- eine Route zum öffentlichen WAN-Netz des jeweils anderen Routers über den ISP
- eine Route zum entfernten LAN-Netz über das Tunnel-Interface

Im nächsten Schritt wurde auf beiden Routern das Tunnel-Interface (Tunnel0) konfiguriert.

Das Tunnelnetz gemäß KN=18 lautet 192.168.18.0/30:

- RA Tunnel0 → 192.168.18.1
- RB Tunnel0 → 192.168.18.2

Als Tunnel-Source wurde jeweils das WAN-Interface (G0/1) verwendet, als Tunnel-Destination die öffentliche IP-Adresse des Gegenrouters.

Das Tunnel-Interface wurde aktiviert und zeigte anschließend den Status up/up.

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no ip domain-lookup
Router(config)#hostname RA
RA(config)#enable secret class
RA(config)#service password-encryption
RA(config)#line con 0
RA(config-line)#password cisco
RA(config-line)#login
RA(config-line)#exit
RA(config)#line vty 0 4
RA(config-line)#password cisco
RA(config-line)#login
RA(config-line)#exit
RA(config)#interface g0/0
RA(config-if)# ip address 10.16.1.1 255.255.255.0
RA(config-if)# no shutdown

RA(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

RA(config)#interface g0/1
RA(config-if)# ip address 203.0.113.2 255.255.255.252
RA(config-if)# no shutdown

RA(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

RA(config)#ip route 198.51.100.12 255.255.255.252 203.0.113.1
RA(config)#

RA(config)#
RA(config)#interface tunnel0
RA(config-if)# ip address 192.168.18.1 255.255.255.252
RA(config-if)# tunnel source g0/1
RA(config-if)# tunnel destination 198.51.100.14
RA(config-if)# no shutdown
RA(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

RA(config-if)#
RA(config-if)#

RA(config)#ip route 10.16.2.0 255.255.255.0 192.168.18.2
RA(config)#end
RA#
%SYS-5-CONFIG_I: Configured from console by console

RA#wr
Building configuration...
[OK]
RA#
```

RB:

```
RB(config)#no ip domain-lookup
RB(config)#enable secret class
RB(config)#service password-encryption
RB(config)#line con 0
RB(config-line)# pass cisco
RB(config-line)# login
RB(config-line)#exit
RB(config)#line vty 0 4
RB(config-line)# pass cisco
RB(config-line)# login
RB(config-line)#exit
RB(config)#
RB(config)#int g0/0
RB(config-if)# ip add 10.16.2.1 255.255.255.0
RB(config-if)# no sh

RB(config-if)#exit
RB(config)#
RB(config)#int g0/1
RB(config-if)# ip add 198.51.100.14 255.255.255.252
RB(config-if)# no sh

RB(config-if)#exit

RB(config)#ip route 203.0.113.0 255.255.255.252 198.51.100.13
RB(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

RB(config)#int tu0

RB(config-if)# ip add 192.168.18.2 255.255.255.252
RB(config-if)# tun s g0/1
RB(config-if)# tun d 203.0.113.2
RB(config-if)# no sh
RB(config-if)#exit

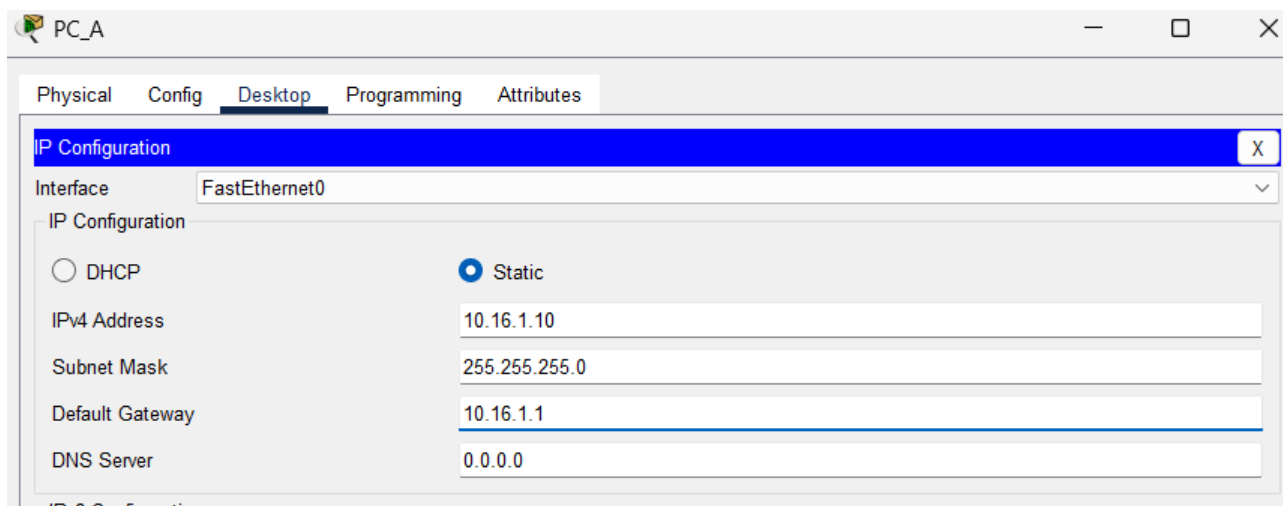
RB(config)#ip route 10.16.1.0 255.255.255.0 192.168.18.1
RB(config)#end
RB#
%SYS-5-CONFIG_I: Configured from console by console

RB#wr
Building configuration...
[OK]
RB#
```

3.Konfiguration der PCs:

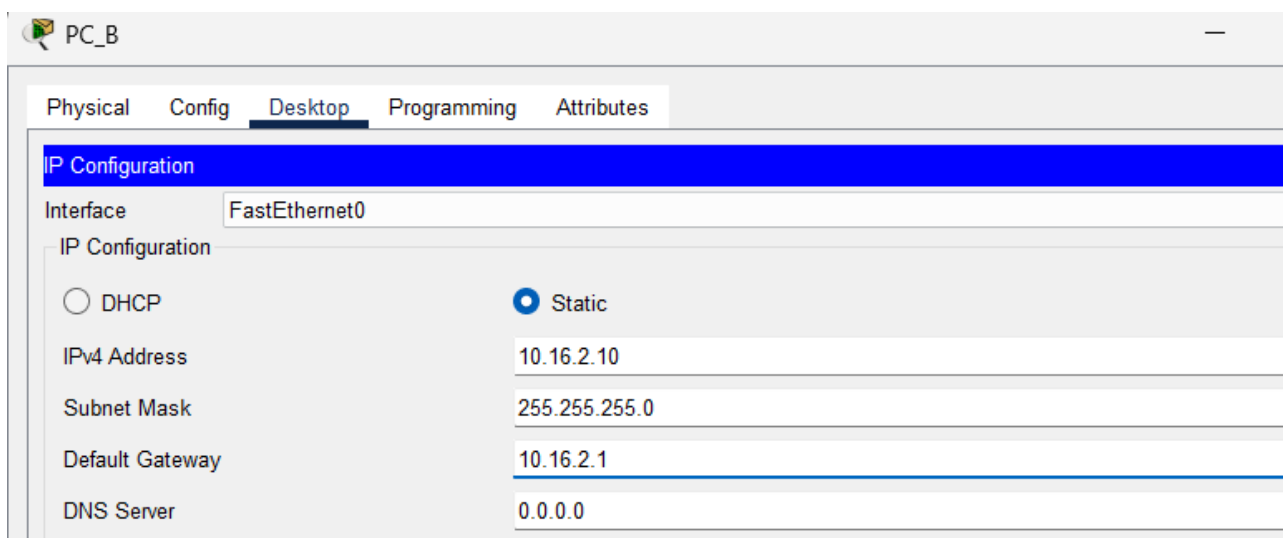
Danach wurden die Endgeräte konfiguriert:

- PC_A → 10.16.1.10/24, Gateway 10.16.1.1
- PC_B → 10.16.2.10/24, Gateway 10.16.2.1



The screenshot shows the configuration window for PC_A. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Static' radio button is chosen. The interface 'FastEthernet0' is selected. The configuration fields are as follows:

Field	Value
IPv4 Address	10.16.1.10
Subnet Mask	255.255.255.0
Default Gateway	10.16.1.1
DNS Server	0.0.0.0



The screenshot shows the configuration window for PC_B. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Static' radio button is chosen. The interface 'FastEthernet0' is selected. The configuration fields are as follows:

Field	Value
IPv4 Address	10.16.2.10
Subnet Mask	255.255.255.0
Default Gateway	10.16.2.1
DNS Server	0.0.0.0

4. Überprüfung der Konfiguration:

Zur Kontrolle wurden auf beiden Routern folgende Befehle ausgeführt:

- show ip interface brief
- show interfaces tunnel0
- show ip route

Dabei wurde geprüft, dass:

- alle Interfaces aktiv sind (up/up)
- das Tunnel-Interface verbunden ist
- die statischen Routen korrekt eingetragen sind

```
RA#show ip interface brief
Interface                IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0       10.16.1.1       YES manual up                  up
GigabitEthernet0/1       203.0.113.2     YES manual up                  up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Serial0/0/0              unassigned      YES unset  administratively down down
Serial0/0/1              unassigned      YES unset  administratively down down
Tunnel0                  192.168.18.1    YES manual up                  up
Vlan1                    unassigned      YES unset  administratively down down
RA#
RA#show interfaces tunnel0
Tunnel0 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 192.168.18.1/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 203.0.113.2 (GigabitEthernet0/1), destination 198.51.100.14
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
      0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out
```

```

RA#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.16.1.0/24 is directly connected, GigabitEthernet0/0
L       10.16.1.1/32 is directly connected, GigabitEthernet0/0
S       10.16.2.0/24 [1/0] via 192.168.18.2
    192.168.18.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.18.0/30 is directly connected, Tunnel0
L       192.168.18.1/32 is directly connected, Tunnel0
    198.51.100.0/30 is subnetted, 1 subnets
S       198.51.100.12/30 [1/0] via 203.0.113.1
    203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/1
L       203.0.113.2/32 is directly connected, GigabitEthernet0/1

```



```

RB#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       10.16.2.1       YES manual up          up
GigabitEthernet0/1       198.51.100.14   YES manual up          up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Serial0/0/0              unassigned      YES unset  administratively down down
Serial0/0/1              unassigned      YES unset  administratively down down
Tunnel0                  192.168.18.2    YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
RB#
RB#show interfaces tunnel0
Tunnel0 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 192.168.18.2/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 198.51.100.14 (GigabitEthernet0/1), destination 203.0.113.2
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out

```

```

RB#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

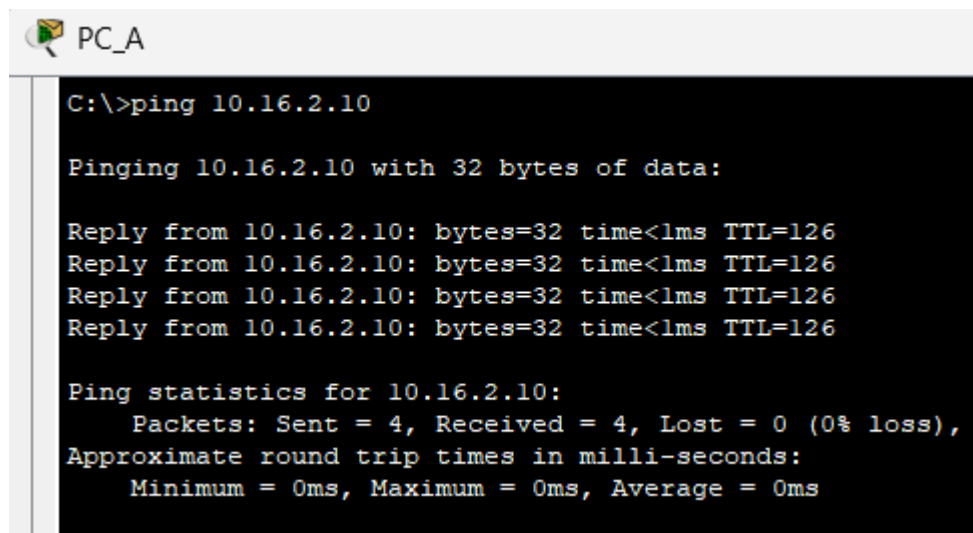
    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
S       10.16.1.0/24 [1/0] via 192.168.18.1
C       10.16.2.0/24 is directly connected, GigabitEthernet0/0
L       10.16.2.1/32 is directly connected, GigabitEthernet0/0
    192.168.18.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.18.0/30 is directly connected, Tunnel0
L       192.168.18.2/32 is directly connected, Tunnel0
    198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.12/30 is directly connected, GigabitEthernet0/1
L       198.51.100.14/32 is directly connected, GigabitEthernet0/1
    203.0.113.0/30 is subnetted, 1 subnets
S       203.0.113.0/30 [1/0] via 198.51.100.13

```

5.End-to-End Test (Ping)

Von PC_A wurde PC_B angepingt:

- ping 10.16.2.10



```

PC_A
C:\>ping 10.16.2.10

Pinging 10.16.2.10 with 32 bytes of data:

Reply from 10.16.2.10: bytes=32 time<1ms TTL=126
Reply from 10.16.2.10: bytes=32 time<1ms TTL=126
Reply from 10.16.2.10: bytes=32 time<1ms TTL=126
Reply from 10.16.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 10.16.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

6. Traceroute (laut Angabe auf RA)

Auf RA wurden zwei Traceroutes durchgeführt:

1. traceroute 10.16.2.1
2. Ergebnis: Hop 1 = 192.168.18.2 (RB Tunnel0)

traceroute 10.16.2.10

Ergebnis:

- Hop 1 = 192.168.18.2
- Hop 2 = 10.16.2.10

Interpretation (kurz):

Der erste Hop ist die Tunneladresse von RB → dadurch ist nachgewiesen, dass der Verkehr über den GRE-Tunnel läuft.

```
RA#traceroute 10.16.2.1
Type escape sequence to abort.
Tracing the route to 10.16.2.1

 1  192.168.18.2    0 msec    1 msec    0 msec
RA#
```

```
RA#traceroute 10.16.2.10
Type escape sequence to abort.
Tracing the route to 10.16.2.10

 1  192.168.18.2    0 msec    0 msec    1 msec
 2  10.16.2.10      0 msec    0 msec    1 msec
RA#
```